2.7.4.2 For each subject whose first attemp t exceeds 1 min or is incomplete, after demonstration or instruction to familiarize the subject with the donning procedure, the test subject should then don the lifejacket without assistance while wearing normal clothing, using the same instruction and timing method as in 2.7.4.1. Each subject should correctly don the lifejacket within a period of 1 min.

Heavy-weather clothing test

2.7.4.3 Each subject should then don the lif ejacket without assistance while wearing heavy-weather clothing, using the same instruction and timing method as in 2.7.4.1. Each subject should don the lifejacket correctly within a period of 1 min.

2.8 Water performance tests

2.8.1 This portion of the test is intended to determine the ability of the lifejacket to assist a helpless person or one in an exhausted or unconscious state and to show that the lifejacket does not unduly restrict movement. The in-water performance of a lifejacket is

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SOURCE: MSC.200(80).pdf
assist a helpless person or one in an exhausted or unconscious state and to show that the lifejacket does not unduly restrict movement. The in-water performance of a lifejacket is evaluated by comparison to the performan ce of a suitable size standard reference lifejacket, i.e. Reference Test Device (RTD) as specified in appendices 1 to 3. All tests should be carried out in fresh water under still conditions.

Test subjects

2.8.2 These tests should be carried out with at least 12 persons as described in 2.7.2. Only good swimmers should be used, since the ability to relax in the water is rarely otherwise obtained.


Clothing

2.8.3 Subjects should wear only swimming costumes.

Preparation for water performance tests

2.8.4 The test subjects should be made familiar with each of the tests set out below, particularly the requirement regarding rela xing and exhaling in the face-down position.

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SOURCE: MSC.200(80).pdf

Preparation for water performance tests

2.8.4 The test subjects should be made familiar with each of the tests set out below, particularly the requirement regarding rela xing and exhaling in the face-down position. The test subject should don the lifejacket, unassisted, using only the instructions provided by the manufacturer. After entering the water, care should be ta ken to ensure that there is no significant amount of air unintentionally trapped in th e lifejacket or swimming costume.

Righting tests

2.8.5 Each test subject should assume a prone, face down position in the water, but with the head lifted up so the mouth is out of the water. The subject's feet should be supported, shoulder width apart, with the heels just below the surface of the water. After assuming a starting position with the legs straight and arms along the sides, the subject should then be instructed in the followi ng sequence to allow the body to gradually and completely relax into a natural floating posture : allow the arms and shoulders to relax; allow the legs to relax; and then the spine a nd neck, letting the head fall into the water

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SOURCE: MSC.200(80).pdf
completely relax into a natural floating posture : allow the arms and shoulders to relax; allow the legs to relax; and then the spine a nd neck, letting the head fall into the water while breathing out normally. During th e relaxation phase, the subject should be maintained in a stable face down position. Im mediately after the subject has relaxed with the face in the water, simulating a state of u tter exhaustion, the subject's feet should be released. The period of time until the mouth of the test subject comes clear of the water should be recorded to the near est 1/10 of a second, starting fr om when the subject's feet are released. The above test should be conduc ted for a total of six times, and the highest and lowest times discarded. The test should then be conducted for a total of six times in the RTD and the highest and lowest times discarded.

Static balance measurements

2.8.6 At the conclusion of the righting tests without making any adjustments in body or lifejacket position, measurements should be made with th e subject floating in the relaxed face-up position of static balance resulting from the preceding tests. The following

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SOURCE: MSC.200(80).pdf
lifejacket position, measurements should be made with th e subject floating in the relaxed face-up position of static balance resulting from the preceding tests. The following measurements should be made (see figure 4):

.1 freeboard the distance measured perpendi cularly from the surface of the water to the lowest point of the subject's mouth where respiration may be impeded, if the mouth were not held shut. The lowest side of the mouth should be measured if the left and right sides are not level;

.2 faceplane angle the angle, relative to the surface of the water, of the plane formed between the most forward part of the forehead and the chin;

.3 torso angle  the angle, relative to vertical, of the line formed by the forward points of the shoulder and hipbone (ilium portion of the pelvis); and

.4 list angle  the angle relative to the surface of the water and a line between the left and right shoulder or a line through the ears if only the head is tilted.
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Figure 4 – Static balance measurements

Assessment

2.8.7 After the water tests descri bed in 2.8.5 and .6 above:

.1 Turning time:  The average turn time for all subjects in the candidate lifejacket should not exceed the aver age time in the RTD, and the number of “no-turns”, if any, should not exceed the number in the RTD;

.2 Freeboard:  The average freeboard of all the subjects should not be less than the average for the RTD;

.3 Torso angles:  The average of all subjects’ tors o angles should be not less than the average for the RTD minus 5°;

.4 Faceplane (head) angles:  The average of all subjects’ faceplane angles should be not less than the average for the RTD minus 5°;

.5 Lifejacket light location:  The position of the lifejacket light should permit it to be visible ove r as great a segment of th e upper hemisphere as is practicable.

Jump and drop tests

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SOURCE: MSC.200(80).pdf
.5 Lifejacket light location:  The position of the lifejacket light should permit

it to be visible ove r as great a segment of th e upper hemisphere as is practicable.

Jump and drop tests

2.8.8 Without readjusting the lifejacket, the test subject should jump vertically into the water, feet first, from a height of at least 1 m while holding the arms over the head. Upon entering the water, the test subject sh ould relax to simulate a state of utter exhaustion. The freeboard to the mouth should be recorded after the test subject comes to rest. The test should be repeated from a hei ght of at least 4.5 m but, when jumping into

Torso angle
degrees from
vertical
Faceplane –
degrees from
horizontalFreeboard mm